Princess Sumaya University for Technology
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Electromagnetics I
Solved Example on E and V

For a spherical charge distribution:

$$\rho_v = \begin{cases} \rho_o(a^2 - r^2), & r < a \\ 0, & r > a \end{cases}$$

a. Find $\boldsymbol{E}$ and V for $r \geq a$.
b. Find $\boldsymbol{E}$ and V for $r \leq a$.
c. Find the total charge.
d. Show that $\boldsymbol{E}$ is maximum when $r = 0.745\, a$.

Solution

For $r \geq a$

$$Q_{enc} = \int \rho_v\, dv = \iiint \rho_0(a^2 - r^2) r^2 \sin\theta \; d\theta\, d\phi\, dr$$

$$Q_{enc} = \rho_0 \int_0^{\pi} \sin\theta\, d\theta \int_0^{2\pi} d\phi \int_0^{a} (a^2 r^2 - r^4)\, dr$$

$$Q_{enc} = 4\pi\rho_0 \left(a^2 \frac{r^3}{3} - \frac{r^5}{5} \right)\Bigg|_0^a = \frac{8\pi}{15}\rho_0$$

$$\Psi = \int \boldsymbol{D}.d\boldsymbol{S} = \varepsilon_o E_r (4\pi r^2)$$

$$\Psi = Q_{enc}$$

$$\varepsilon_o E_r (4\pi r^2) = \frac{8\pi}{15}\rho_0$$

$$E_r = \frac{2\rho_0}{15\varepsilon_o r^2}$$

$$\boldsymbol{E} = \frac{2\rho_0}{15\varepsilon_o r^2}\boldsymbol{a_r}$$

$$V = -\int \boldsymbol{E}.d\boldsymbol{l} = -\frac{2\rho_0}{15\varepsilon_o}\int r^{-2}.dr = \frac{2\rho_0}{15\varepsilon_o r} + C_1$$

Since

$$V(r^- > 0) = 0$$

$$C_1 = 0$$

$$V = \frac{2\ \rho_0}{15\ \varepsilon_o r}$$

b. For $r \leq a$

$$Q_{enc} = \rho_0(4\pi)\left(a^2\frac{r^3}{3} - \frac{r^5}{5}\right)\Bigg|_0^r = 4\pi\rho_0\left(a^2\frac{r^3}{3} - \frac{r^5}{5}\right)$$

$$E_r = \frac{Q_{enc}}{4\pi\varepsilon_o r^2} = \frac{\rho_0}{\varepsilon_o}\left(a^2\frac{\mathrm{r}}{3} - \frac{r^3}{5}\right)$$

$$\boldsymbol{E} = \frac{\rho_0}{\varepsilon_o}\left(a^2\frac{r}{3} - \frac{r^3}{5}\right)\boldsymbol{a}_r$$

$$V = -\int \boldsymbol{E}.d\boldsymbol{l} = -\frac{\rho_0}{\varepsilon_o}\left(a^2\frac{r^2}{6} - \frac{r^4}{20}\right) + C_2$$

$$V = \frac{\rho_0}{\varepsilon_o}\left(\frac{r^4}{20} - a^2\frac{r^2}{6}\right) + C_2$$

Since $V(r = a^+) = V(r = a^-)$:

$$\frac{2\rho_0}{15\varepsilon_o a} = \frac{\rho_0}{\varepsilon_o}\left(\frac{a^4}{20} - \frac{a^4}{6}\right) + C_2$$

$$C_2 = \frac{2\rho_0}{15\varepsilon_o a} + \frac{7\rho_0 a^4}{60\varepsilon_o}$$

$$V = \frac{\rho_0}{\varepsilon_o}\left(\frac{r^4}{20} - a^2\frac{r^2}{6}\right) + \frac{2\rho_0}{15\varepsilon_o a} + \frac{7\rho_0 a^4}{60\varepsilon_o}$$

c. The total charge is found in part (a) as:

$$Q = \frac{8\pi}{15}\rho_0$$

d. For $r \geq a$, $\boldsymbol{E}$ decays to zero with no maxima. For $r \leq a$:

$$E_r = \frac{\rho_0}{\varepsilon_o}\left(a^2 \frac{\mathrm{r}}{3} - \frac{r^3}{5}\right)$$

$$\frac{\partial E_r}{\partial r} = \frac{\rho_0}{\varepsilon_o}\left(\frac{a^2}{3} - \frac{3r^2}{5}\right) = 0$$

$$r = \frac{a\sqrt{5}}{3}$$

hence

$$r = 0.7453\ a$$